

Deepak Mallampati

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Kent, OH (US, F-1 OPT) -- open to EU remote / EOR / sponsorship

PROFESSIONAL SUMMARY

Applied AI / LLM engineer (4+ years) shipping production **RAG, agentic, and LLMOps** systems. Core strength is exactly what this role centers on: **context engineering** -- designing retrieval strategies, prompt structures, and input pipelines that maximize LLM output quality. At Progress Solutions: high-performance RAG pipelines (dense retrieval + cross-encoder re-ranking, ~35% precision lift, 40% hallucination reduction) and a conductor-subagent **agentic** framework (42% token reduction). Built a structured **LLM evaluation framework** (RAGAS, BERTScore, LLM-as-a-judge style scoring) with latency/accuracy dashboards. Translation-domain proof point: shipped MangaLens, a real-time AI translation product live across 29+ sites. Strong Python production discipline; comfortable moving fluidly between ML experimentation and production engineering.

CORE COMPETENCIES

Context Engineering (retrieval + prompt structures) . Production RAG (dense + cross-encoder re-rank) . Agentic Workflows (orchestration, self-correction) . LLMOps (deploy, monitor, iterate) . LLM Evaluation (RAGAS, BERTScore, LLM-as-a-judge) . Python production systems . Vector DBs (FAISS, Pinecone, Weaviate) . Machine Translation tooling (MangaLens) . Experiment design (baselines, measurable outcomes)

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

AI Engineer -- USA | Healthcare Technology, AI Consulting

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents -- reduced token consumption **42%** while sustaining task accuracy across multi-step workflows.
- Engineered **retrieval-augmented generation (RAG) pipelines** over large-scale healthcare corpora; dense vector retrieval + cross-encoder re-rank lifted precision ~35% and materially reduced hallucinations on domain queries.
- Built **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, Laplace differential privacy) enabling compliant analytics with zero PII exposure.
- Packaged LLM/ML inference as **FastAPI/Flask** services containerized with Docker; structured logging + load simulation cut post-deployment defects ~30%.
- Established structured **LLM evaluation framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to catch regressions pre-deployment.

Vanguard

Jan 2024 -- Jan 2025

AI Engineer Intern -- USA | Financial Services

- Developed **25+ AI agents** and backend services for Vanguard's internal AI platform; integrated with production data pipelines + observability across 20+ internal teams.
- Benchmarked **GPT-4o, Claude Sonnet, Gemini** against internal eval sets; improved task success +27% and informed platform model-routing decisions.
- Engineered safeguards and content controls for policy-compliant AI; reduced unsafe outputs **42%** while preserving response quality.
- Shipped **12 APIs and microservices**; cut p95 latency 1.5s -> **800ms** and accelerated feature integration **40%**.

Kent State University

Oct 2023 -- Jan 2024

ML & Gen AI Research Assistant -- USA | Privacy-preserving ML

- Built privacy-preserving ML pipeline on clinical hospital-readmission dataset; reduced re-identification risk **3.38%** -> **0.32%** while retaining 99% predictive performance.
- Designed composite privacy-utility scoring framework benchmarking six anonymization configs; quantified DP utility cliff at low epsilon.
- Fine-tuned **Qwen3-27B** via 4-bit QLoRA on H100 GPU for controllable long-form generation. Authored IEEE-format paper.

Emerson

Jun 2022 -- Apr 2023

ML Trainee -- India | Industrial Automation

- Trained supervised regression + classification models on operational pipeline-storage data for equipment-failure prediction and anomaly detection.
- Fine-tuned **BERT/RoBERTa** for NER + text classification, achieving **89% F1** on custom domain entity extraction.

Suvidha Foundation

Jan 2022 -- Mar 2022

ML Intern -- India | Nonprofit Tech

- Built end-to-end video summarization + highlight-generation pipeline; processed 5,000+ videos; reduced manual editing **70%**.

PROJECTS

MangaLens shipped -- Chrome Web Store

Real-time AI translation Chrome extension for manga/webtoons across **29+ sites**. Multi-provider AI abstraction (Claude, GPT-4o, Gemini, Ollama). Full ship discipline: privacy policy, scoped permissions, Chrome review approval.

TypeScript, Manifest V3, content scripts, multi-provider abstraction

career-ops

Autonomous AI job-search pipeline scanning listings, scoring fit, generating tailored applications overnight. Compresses hours of repetitive job-hunting into automated workflow.

Node.js, Playwright, Claude API, GitHub Actions, FastAPI

Privacy-Preserving Clinical ML Pipeline

k-anonymity + l-diversity + DP framework analyzing sensitive patient data without exposing identities; quantifies privacy vs utility tradeoff.

Python, scikit-learn, statistical privacy

EDUCATION

M.S. in Computer Science -- Kent State University, Ohio, USA

2025

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite)

AI/LLM & Agentic: LLM APIs (Claude, GPT-4o, Gemini, Ollama), LangChain, LangGraph, RAG, multi-agent workflows, prompt engineering, structured outputs, eval pipelines, guardrails, grounding

Vector Search: FAISS, Pinecone, Weaviate, semantic + hybrid search, cross-encoder re-ranking, chunking strategies

MLOps & Cloud: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, Content Safety), GCP Vertex AI, Docker, Kubernetes, MLflow, observability

Data & Compliance: PostgreSQL, MongoDB, Redis, HIPAA, SOC 2, PCI DSS, PII masking, audit logging

Work Authorization: US-based (F-1 OPT); EU work via EOR/contractor or sponsorship to be confirmed